

Topic: Evolution of Computers — Self-Learning

Q. Trace the evolution of computers through their five generations. Highlight the key technological advancements and characteristics of each generation.

📌 **Definition to Remember**

The **Evolution of Computers** is divided into five "**Generations**", each marked by a major technological breakthrough that made computers smaller, cheaper, faster, and more reliable. The progression moved from Vacuum Tubes → Transistors → Integrated Circuits → Microprocessors → Artificial Intelligence.

Evolution of Computers (Self-Learning)

Generation	Core Technology	Key Characteristics & Examples
First (1940s - 1950s)	Vacuum Tubes (circuitry) & Magnetic Drums (memory).	Massive (room-sized), high heat, high power consumption, very unreliable. Used binary Machine Language. Ex: ENIAC, UNIVAC.
Second (1950s - 1960s)	Transistors	Smaller, faster, more energy-efficient than vacuum tubes. Transitioned to Assembly language and early high-level languages (COBOL, FORTRAN). Ex: IBM 7094.
Third (1960s - 1970s)	Integrated Circuits (ICs)	Miniaturized transistors placed on silicon chips. Introduced keyboards, monitors, and Operating Systems (multi-programming). Ex: IBM 360 series.
Fourth (1970s - Present)	Microprocessors (VLSI)	Thousands of ICs on a single silicon chip (Very Large Scale Integration). Led to personal computers, GUIs, mouse, and networking/Internet. Ex: Apple Mac, IBM PC.
Fifth (Present & Beyond)	Artificial Intelligence (AI)	Based on parallel processing and Machine Learning. Focuses on natural language processing, self-organization, and quantum computing. Ex: Siri, Alexa, IBM Watson.

★ **Must-Write Points (for 10 marks)**

1. Computer evolution is categorized into **Five Generations**.
2. **1st Gen**: Used **Vacuum tubes**; massive, generated high heat, used machine language (ENIAC).
3. **2nd Gen**: Replaced tubes with **Transistors**; smaller/faster, used assembly and high-level languages (COBOL).
4. **3rd Gen**: Used **Integrated Circuits (ICs)** on silicon chips; introduced OS, keyboards, monitors.
5. **4th Gen**: Used **Microprocessors (VLSI)**; brought computers to desktops, introduced networking/Internet.
6. **5th Gen**: Focused on **Artificial Intelligence (AI)** and parallel processing; natural language understanding.
7. The consistent trend across generations: Smaller size, lower cost, higher speed, and greater reliability.

⚡ **Quick Recall**

Generations: 1st (Vacuum Tubes, Room size) → 2nd (Transistors, Assembly Lang) → 3rd (Integrated Circuits, OS) → 4th (Microprocessors/VLSI, PCs/Internet) → 5th (AI, Machine Learning)